Exercise 04: Positional Information

A controlled numerical ablation demonstrating Rotary Position Embedding (RoPE).

Setup

I process the sentence

"I like cats"

with identical content vectors for all three positions:

$$Q = K = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

I hold content constant to ensure any differences in attention scores come purely from position. For this exercise, the projection dimension is $d_k = 2$.

1. Plain Q/K Scores

Without positional information, I compute the raw query-key dot products:

$$S_{\text{plain}} = QK^\top = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

2. Plain Attention

Since all scores are identical, scaling by $\sqrt{2}$ and applying row-wise softmax yields a uniform distribution. The model is completely blind to sequence order:

$$A_{\text{plain}} = \text{softmax}\left(\frac{S_{\text{plain}}}{\sqrt{d_k}}\right) = \begin{bmatrix} 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}.$$

3. RoPE Rotation Definition

RoPE applies a rotation to each row vector q_p and k_p depending on its absolute position index $p \in \{0, 1, 2\}$. This example uses a single frequency $\theta = 1$ radian.

For a given position p, the rotation angle is $\phi_p = p\theta$. The rotated vector is:

$$q_p^R = q_p R(\phi_p)^\top$$

where the rotation matrix is:

$$R(\phi_p) = \begin{bmatrix} \cos(\phi_p) & -\sin(\phi_p) \\ \sin(\phi_p) & \cos(\phi_p) \end{bmatrix}.$$

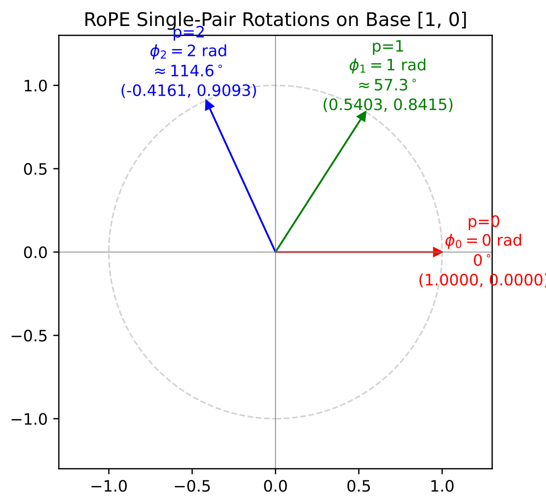
4. Compute Rotation Angles $\phi_p = p\theta$

In Rotary Position Embedding, the absolute sequence index p acts as a scalar multiplier for a base frequency θ. This forces the vector representation to physically rotate by an angle proportional to its sequence position.

Given my controlled sequence length of 3 and a base frequency $\theta = 1$ radian, I compute the discrete rotation angles linearly:

$$\phi_0 = 0 \times 1 = 0 \text{ radians}$$
$$\phi_1 = 1 \times 1 = 1 \text{ radian}$$
$$\phi_2 = 2 \times 1 = 2 \text{ radians}$$

Notice that token 0 (`I`) receives zero rotation, effectively pinning the start of my sequence, while subsequent tokens rotate progressively further around the unit circle.



As shown in the figure above, the absolute position index p determines the rotation angle ϕ_p, which is measured in radians (the degree labels are only an intuitive visual aid). Python dynamically computes the corresponding trigonometric components $\cos(\phi_p)$ and $\sin(\phi_p)$ to plot the exact coordinates. The base vector $[1, 0]$ moves around the unit circle according to these components. Notice that this rotation mathematically preserves the vector's norm (its length stays constant at 1); only its direction changes.

5. Evaluate Rotated Vectors for each p

First, compute the trigonometric values:

$$\begin{aligned} \cos(0) &= 1, & \sin(0) &= 0 \\ \cos(1) &\approx 0.5403, & \sin(1) &\approx 0.8415 \\ \cos(2) &\approx -0.4161, & \sin(2) &\approx 0.9093 \end{aligned}$$

I rotate the row vector $q_p = [1, 0]$ using $q_p^R = q_p R(\phi_p)^\top$.

For $p = 0$:

$$q_0^R = [1, 0] \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}^\top = [1, 0].$$

For $p = 1$:

$$q_1^R = [1, 0] \begin{bmatrix} 0.5403 & -0.8415 \\ 0.8415 & 0.5403 \end{bmatrix}^\top = [1, 0] \begin{bmatrix} 0.5403 & 0.8415 \\ -0.8415 & 0.5403 \end{bmatrix} \approx [0.5403, 0.8415].$$

For $p = 2$:

$$q_2^R = [1, 0] \begin{bmatrix} -0.4161 & -0.9093 \\ 0.9093 & -0.4161 \end{bmatrix}^\top = [1, 0] \begin{bmatrix} -0.4161 & 0.9093 \\ -0.9093 & -0.4161 \end{bmatrix} \approx [-0.4161, 0.9093].$$

6. Construct Rotated Matrices Q_R and K_R

Since Q and K are identical, their rotated versions Q_R and K_R are formed by assembling the stacked vectors q_0^R, q_1^R, q_2^R. In this controlled exercise, because the same position-dependent RoPE rotation is applied to identical base vectors, $q_p^R = k_p^R$ for this specific setup.

I assemble the rotated vectors into matrices:

$$Q_R = \begin{bmatrix} q_0^R \\ q_1^R \\ q_2^R \end{bmatrix} \quad \text{and} \quad K_R = \begin{bmatrix} k_0^R \\ k_1^R \\ k_2^R \end{bmatrix}.$$

In symbolic trigonometric form, this is:

$$Q_R = K_R = \begin{bmatrix} \cos(0\theta) & \sin(0\theta) \\ \cos(1\theta) & \sin(1\theta) \\ \cos(2\theta) & \sin(2\theta) \end{bmatrix}.$$

For $\theta = 1$, the numerical approximation is:

$$Q_R = K_R \approx \begin{bmatrix} 1.0000 & 0.0000 \\ 0.5403 & 0.8415 \\ -0.4161 & 0.9093 \end{bmatrix}.$$

Before computing S_{RoPE}, I also explicitly show $K_R^\top$:

$$K_R^\top = \begin{bmatrix} \cos(0\theta) & \cos(1\theta) & \cos(2\theta) \\ \sin(0\theta) & \sin(1\theta) & \sin(2\theta) \end{bmatrix}$$

and its numerical approximation:

$$K_R^\top \approx \begin{bmatrix} 1.0000 & 0.5403 & -0.4161 \\ 0.0000 & 0.8415 & 0.9093 \end{bmatrix}.$$

This makes the matrix multiplication dimensions visually obvious: Q_R is 3×2, $K_R^\top$ is 2×3, yielding a 3×3 score matrix S_{RoPE}.

7. Compute RoPE Score Matrix S_{RoPE}

The score matrix is defined as $S_{\text{RoPE}} = Q_R K_R^\top$. Explicitly, each entry $S_{ij} = q_i^R \cdot k_j^R$.

Symbolically, the full matrix is:

$$S_{\text{RoPE}} = \begin{bmatrix} q_0^R \cdot k_0^R & q_0^R \cdot k_1^R & q_0^R \cdot k_2^R \\ q_1^R \cdot k_0^R & q_1^R \cdot k_1^R & q_1^R \cdot k_2^R \\ q_2^R \cdot k_0^R & q_2^R \cdot k_1^R & q_2^R \cdot k_2^R \end{bmatrix}.$$

The rows correspond to the rotated queries q_i^R, and the columns correspond to the rotated keys k_j^R. Each cell represents one query-key dot product. I will compute all nine entries using the already-derived numerical rotated vectors.

For S_{00}:

$$S_{00} = (1)(1) + (0)(0) = 1.0000.$$

For S_{01}:

$$S_{01} = (1)(0.5403) + (0)(0.8415) = 0.5403.$$

For S_{02}:

$$S_{02} = (1)(-0.4161) + (0)(0.9093) = -0.4161.$$

For S_{10}:

$$S_{10} = (0.5403)(1) + (0.8415)(0) = 0.5403.$$

For the diagonal entry S_{11}, $q_1^R = k_1^R = [\cos(1), \sin(1)]$. Symbolically:

$$\begin{aligned} S_{11} &= \cos^2(1) + \sin^2(1) \\ &= 1.0000. \end{aligned}$$

This follows directly from the identity $\cos^2(x) + \sin^2(x) = 1$.

For the off-diagonal entry S_{12}, I take the dot product of q_1^R and k_2^R. First, I write their symbolic trigonometric forms:

$$q_1^R = [\cos(1), \sin(1)] \quad \text{and} \quad k_2^R = [\cos(2), \sin(2)].$$

Then I write the dot product symbolically:

$$\begin{aligned} S_{12} &= q_1^R (k_2^R)^\top \\ &= [\cos(1), \sin(1)] \begin{bmatrix} \cos(2) \\ \sin(2) \end{bmatrix} \\ &= \cos(1)\cos(2) + \sin(1)\sin(2). \end{aligned}$$

Only now do I substitute the rounded numerical values:

$$\begin{aligned} &\cos(1) \approx 0.5403, && \sin(1) \approx 0.8415 \\ &\cos(2) \approx -0.4161, && \sin(2) \approx 0.9093 \end{aligned}$$

Therefore:

$$\begin{aligned} S_{12} &\approx (0.5403)(-0.4161) + (0.8415)(0.9093) \\ &\approx -0.2248 + 0.7651 \\ &\approx 0.5403. \end{aligned}$$

Notice that the same result follows immediately from the trigonometric difference identity $\cos(a - b) = \cos(a)\cos(b) + \sin(a)\sin(b)$:

$$\begin{aligned} S_{12} &= \cos(1-2) \\ &= \cos(-1) \\ &= \cos(1) \\ &\approx 0.5403. \end{aligned}$$

For S_{20}:

$$S_{20} = (-0.4161)(1) + (0.9093)(0) = -0.4161.$$

For S_{21}, first symbolically:

$$S_{21} = \cos(2)\cos(1) + \sin(2)\sin(1).$$

Then numerically:

$$\begin{aligned} S_{21} &\approx (-0.4161)(0.5403) + (0.9093)(0.8415) \\ &\approx 0.5403. \end{aligned}$$

For S_{22}, first symbolically:

$$S_{22} = \cos^2(2) + \sin^2(2) = 1.0000.$$

Then optionally show the decimal check:

$$S_{22} \approx (-0.4161)^2 + (0.9093)^2 \approx 1.0000.$$

After computing each query-key dot product, I assemble the nine values into the 3×3 RoPE score matrix:

$$S_{\text{RoPE}} \approx \begin{bmatrix} 1.0000 & 0.5403 & -0.4161 \\ 0.5403 & 1.0000 & 0.5403 \\ -0.4161 & 0.5403 & 1.0000 \end{bmatrix}.$$

The diagonal entries are S_{00}, S_{11}, and S_{22}. Each diagonal entry compares a rotated query with the rotated key at the SAME position. In this controlled setup, $q_p^R = k_p^R$ and each vector has unit norm. Therefore:

$$S_{pp} = q_p^R \cdot k_p^R = \|q_p^R\|^2 = 1.0000.$$

Important Note: This diagonal equality is specific to this controlled setup where $Q = K$ and the base vectors are identical unit vectors. It does not imply that diagonal attention scores are generally equal to 1 in a real attention layer.

Because $Q_R = K_R$ in this controlled example, $S_{\text{RoPE}} = Q_R Q_R^\top$, so the resulting matrix is symmetric ($S_{ij} = S_{ji}$).

8. Compare Entries with Equal Relative Offsets

Notice the strong relative-position structure in S_{RoPE}. Entries corresponding to token pairs with a relative sequence offset of 1 (e.g., $S[0,1]$ and $S[1,2]$) are exactly equal (≈ 0.5403). The entry with an offset of 2 ($S[0,2]$) drops to ≈ -0.4161.

9. Relative-Position Identity

Why does this relative structure emerge from absolute rotations? Consider a query at position m and a key at position n:

$$q_m^R = q_m R(m\theta)^\top, \qquad k_n^R = k_n R(n\theta)^\top.$$

The dot product starts by substituting the definitions $q_m^R = q_m R(m\theta)^\top$ and $k_n^R = k_n R(n\theta)^\top$:

$$q_m^R (k_n^R)^\top = q_m R(m\theta)^\top \left(k_n R(n\theta)^\top\right)^\top.$$

Using the transpose rule $(AB)^\top = B^\top A^\top$, I have:

$$\left(k_n R(n\theta)^\top\right)^\top = R(n\theta) k_n^\top.$$

Therefore:

$$q_m^R (k_n^R)^\top = q_m R(m\theta)^\top R(n\theta) k_n^\top.$$

Using the geometric property of 2D rotation matrices $R(a)^\top R(b) = R(b - a)$, I get:

$$q_m^R (k_n^R)^\top = q_m R((n - m)\theta) k_n^\top.$$

Because my base vectors are identically $q_m = k_n = [1, 0]$, this simplifies to $\cos((n - m)\theta)$.

- Offset 0: $\cos(0) = 1.0000$
- Offset ± 1: $\cos(1) \approx 0.5403$
- Offset ± 2: $\cos(2) \approx -0.4161$

This proves that absolute-position-dependent rotations create relative-position dependence in the dot product.

A Trigonometric View of the Same Identity

I can also derive the same result explicitly using trigonometry. Starting from my controlled setup $q_m = k_n = [1, 0]$, the vectors after RoPE are:

$$q_m^R = [\cos(m\theta), \sin(m\theta)] \quad \text{and} \quad k_n^R = [\cos(n\theta), \sin(n\theta)].$$

Then derive the dot product explicitly:

$$\begin{aligned} S_{mn} &= q_m^R (k_n^R)^\top \\ &= \cos(m\theta)\cos(n\theta) + \sin(m\theta)\sin(n\theta) \end{aligned}$$

Using the standard trigonometric difference identity $\cos(a - b) = \cos(a)\cos(b) + \sin(a)\sin(b)$, I get:

$$S_{mn} = \cos(m\theta - n\theta) = \cos((m - n)\theta).$$

Since the cosine function is even ($\cos(-x) = \cos(x)$), this is equivalent to:

$$\boxed{S_{mn} = \cos((n - m)\theta)}$$

This explicitly validates my hand-calculated $S_{12} \approx 0.5403$ from Section 7, since $\cos((2 - 1) \times 1) = \cos(1) \approx 0.5403$.

Important Warning: This simplified cosine-only formula $S_{mn} = \cos((n - m)\theta)$ is exactly true *only* for this specific controlled setup where the base Q/K vectors are identical unit vectors $[1, 0]$. The general RoPE attention formula for arbitrary content vectors remains the full matrix identity $q_m^R (k_n^R)^\top = q_m R((n - m)\theta) k_n^\top$.

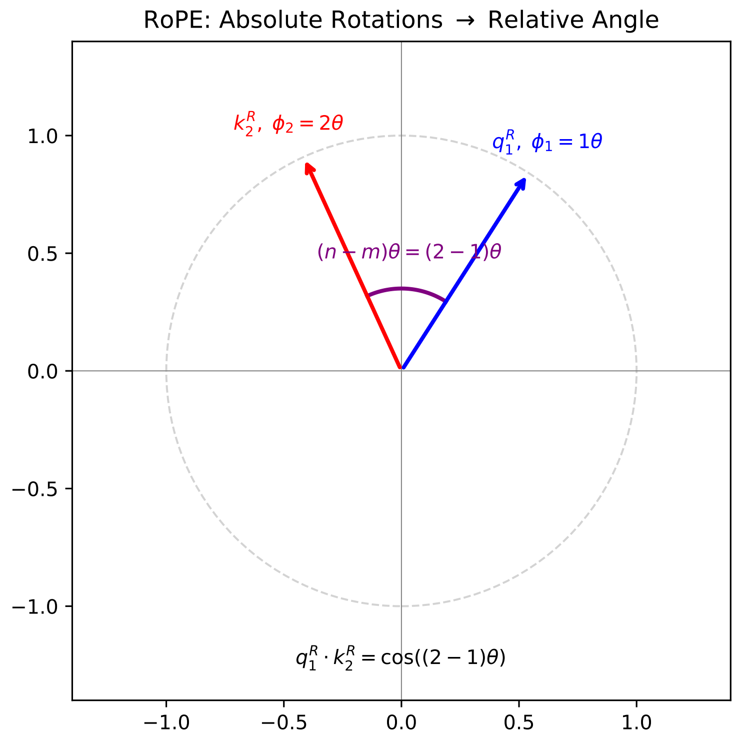


Figure 1: The query and key are rotated by absolute angles $m\theta$ and $n\theta$. Their angular difference is $(n-m)\theta$, which is the relative position signal exposed by the dot product in this controlled unit-vector example.

10. Scale by $\sqrt{d_k}$

I scale the raw scores by $\sqrt{d_k} = \sqrt{2} \approx 1.4142$. For example:

$$1/\sqrt{2} \approx 0.7071$$
$$0.5403/\sqrt{2} \approx 0.3821$$
$$-0.4161/\sqrt{2} \approx -0.2943$$

Applying this to the full matrix:

$$S_{\text{scaled}} = \frac{S_{\text{RoPE}}}{\sqrt{d_k}} \approx \begin{bmatrix} 0.7071 & 0.3821 & -0.2943 \\ 0.3821 & 0.7071 & 0.3821 \\ -0.2943 & 0.3821 & 0.7071 \end{bmatrix}.$$

11. Softmax

Applying row-wise softmax. Let's compute the first row explicitly:

$$S_{\text{scaled}}[0] \approx [0.7071, 0.3821, -0.2943]$$

First, I exponentiate each value:

$$\begin{aligned}\exp(0.7071) &\approx 2.0281\\ \exp(0.3821) &\approx 1.4654\\ \exp(-0.2943) &\approx 0.7451\end{aligned}$$

The normalization constant Z is the sum of these exponentiated values:

$$Z \approx 2.0281 + 1.4654 + 0.7451 \approx 4.2386$$

Dividing each exponentiated value by Z gives the attention weights for the first row:

$$\begin{aligned}A_{\text{RoPE}}[0] &\approx \left[\frac{2.0281}{4.2386}, \frac{1.4654}{4.2386}, \frac{0.7451}{4.2386}\right]\\ &\approx [0.4785, 0.3457, 0.1758]\end{aligned}$$

For the middle row, $S_{\text{scaled}}[1] \approx [0.3821, 0.7071, 0.3821]$:

$$\begin{aligned}\exp(0.3821) &\approx 1.4654\\ \exp(0.7071) &\approx 2.0281\\ Z &\approx 1.4654 + 2.0281 + 1.4654 \approx 4.9589\end{aligned}$$

$$A_{\text{RoPE}}[1] \approx \left[\frac{1.4654}{4.9589}, \frac{2.0281}{4.9589}, \frac{1.4654}{4.9589}\right] \approx [0.2955, 0.4090, 0.2955]$$

Using symmetry for the third row, I assemble the final attention matrix:

$$A_{\text{RoPE}} \approx \begin{bmatrix} 0.4785 & 0.3457 & 0.1758\\ 0.2955 & 0.4090 & 0.2955\\ 0.1758 & 0.3457 & 0.4785 \end{bmatrix}.$$

I can verify that each row sums to approximately 1.

12. Compare Before vs After

Without position:

$$A_{\text{plain}}[0] = [0.3333, 0.3333, 0.3333]$$

With position (RoPE):

$$A_{\text{RoPE}}[0] \approx [0.4785, 0.3457, 0.1758]$$

In this controlled example, RoPE changes the attention weights according to relative position, allowing identical content vectors at different positions to receive different compatibility scores.

13. Final Conceptual Answer

RoPE uses absolute position indices to rotate Q and K. Why can the resulting query-key dot product nevertheless depend on relative position?

Because a dot product between two vectors measures the angle between them. If vector A is rotated by angle α and vector B is rotated by angle β, the angle between their rotated forms only changes by the relative difference $(\beta - \alpha)$.

14. Bonus: Norm Preservation

Why does rotating both Q and K preserve their individual vector norms?

Rotation matrices are orthogonal ($R^\top R = I$). Therefore, the length of any vector multiplied by R remains unchanged. RoPE only changes the angles between different vectors, never their individual magnitudes.

Pipeline Summary

$$\boxed{\begin{array}{c} Q, K \longrightarrow \text{apply } R(p\theta) \text{ to position } p \longrightarrow Q_R, K_R \\ \longrightarrow \text{Scores} = Q_R K_R^\top \longrightarrow \text{Scale by } \sqrt{d_k} \longrightarrow \text{softmax} \longrightarrow A_{\text{RoPE}} \end{array}}$$

RoPE applies absolute-position-dependent rotations to Q and K, while their dot product exposes relative positional structure.

Numerical Verification

The companion `answer_key.py` reproduces these matrices using basic PyTorch operations and checks the results with assertions.